Precision, Simplicity, Appeal: Decoding the impact of visual design in public service announcements

Alice Mao

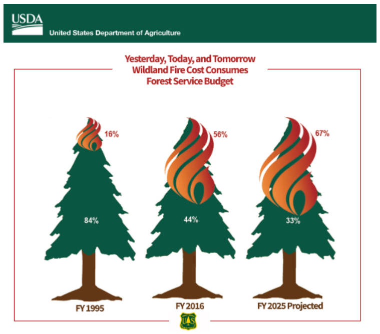

Fig. 1: Original infographic

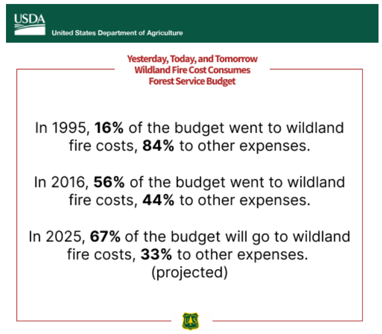

Fig. 2: Only informative text

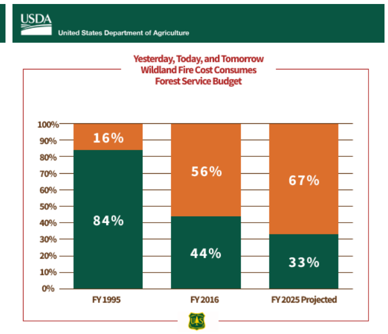

Fig. 3: Traditional graph

Fig. 4: Five different formats of the same infographic from the United States Department of Agriculture

Abstract—Public service announcements (PSAs) often rely on visual design to engage audiences and communicate critical information. This study investigates how three common PSA formats — infographics, traditional graphs, and text-only representations — affect comprehension, memory, and user preferences. Participants viewed recreated versions of real government-issued PSAs, one of each format per topic, and responded to comprehension, recall, and ranking tasks. Results showed no significant difference in comprehension across formats overall, though topic-specific effects suggest that visual clarity can influence understanding. Memory performance declined most for graphs. While infographics were consistently rated highest for overall preference, attention-grabbing, raising awareness, and memorability, graphs were perceived as the most trustworthy. These findings highlight the importance of balancing engagement and clarity in infographic PSAs.

Index Terms—Visualization

1 Introduction

Public Service Announcements (PSAs) are strategically designed messages that aim to raise public awareness, educate communities, and promote beneficial programs, services, or behaviors. PSAs serve as a vital communication tool for governmental agencies, health organizations, and advocacy groups, particularly when they aim to quickly and persuasively spread important information to a broad audience. These messages are distributed across a range of formats, such as billboards, social media posts, videos, infographics, and plain text, and are often grounded in data and research to enhance their credibility and persuasion.

One popular type of PSA is an infographic, which can condense complex data into visually engaging, easily digestible formats. Infographic PSAs often incorporate a blend of textual annotations, numerical data, and visual metaphors to inform and attract attention simultaneously. These visual embellishments can enhance emotional engagement, make key messages more salient, and improve information retention [1]. However, integrating design features also introduces the potential for distraction, misinterpretation, or cognitive overload, especially for viewers unfamiliar with reading charts or interpreting visual cues.

This balance between aesthetic appeal and informational clarity is particularly important in public communication, where misunderstanding a message can have real-world consequences. A poorly designed or overly complex PSA can confuse its intended audience, diminishing its effectiveness and, in some cases, spreading misinformation. Conversely, a well-designed PSA that balances creativity with clarity can significantly enhance public understanding and motivate action. Thus, it is essential to understand how design choices influence interpretation, especially in contexts where data-driven visuals play a central role in conveying public health or safety information.

This leads to an important question: Does making eye-catching, creative infographic PSAs affect how we understand them? The present study seeks to explore this question, particularly for infographic PSAs incorporating data visualizations. To investigate this, we developed three experimental conditions for each PSA topic: (1) a plain statistical chart, (2) an infographic with visual embellishments, and (3) a text-only description of the same information. Participants viewed one randomly assigned version of each PSA design and were assessed on their comprehension, recall, and preference based on five different definitions of preference.

The findings of this study aim to shed light on which visualization formats most effectively enhance the communication of PSA messages. These insights carry practical implications for public health campaigns, government messaging, and other informative efforts that seek to maximize the clarity and impact of their communications. Furthermore, this research contributes to broader discussions within the visualization community regarding the influence of visual embellishments on the comprehension and memorability of data.

- *Josiah Carberry is with Brown University. E-mail: jcarberry@example.com*
- *Ed Grimley is with Grimley Widgets, Inc. E-mail: ed.grimley@example.com.*

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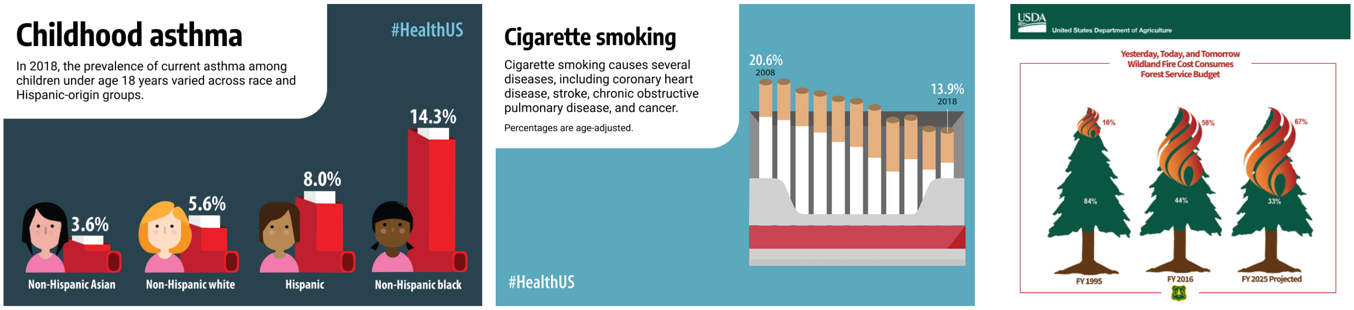
2 Related Work

Visual Embellishment

Data visualization design has long been shaped by competing philosophies regarding the role of aesthetics and simplicity. One school of thought advocates for visual embellishment, which involves incor-



(a) Asthma PSA [2]

(b) Cigarette PSA [2]

(c) Fire Safety PSA [6]

Fig. 5: Original infographic PSAs on asthma, smoking, and fire safety.

porating imagery, illustrations, and thematic elements into charts to emphasize patterns, evoke emotional responses, and improve memorability [1]. In contrast, minimalist designers argue that charts should remain clean, stripped of non-essential elements, to enhance clarity and reduce distractions. Proponents of minimalism, such as Edward Tufte, contend that eliminating decorative elements decreases cognitive load and reduces the likelihood of misinterpretation by allowing viewers to focus solely on the data [1].

Bateman et al. [1] investigated the effects of visual embellishments on comprehension, recall, and preference by comparing plain charts with highly stylized infographics. Their findings revealed that embellishments did not hinder comprehension and significantly improved long-term recall. Additionally, participants expressed a strong preference for visually embellished charts. These results suggest that embellishments may enhance memorability by strengthening associations between data and imagery. However, the authors also caution that such enhancements could introduce unintended biases due to the additional meaning conveyed by decorative elements.

This debate between aesthetic engagement and interpretive clarity is central to the present study, which systematically compares embellished infographic PSAs with minimalist chart-based versions to examine how visual design choices influence comprehension, memory, and viewer preference.

An argument for a text-only variant

Other research has explored the role of text in data visualization, specifically how a combination of textual and graphical representation influences comprehension, recall, and user preference. Stokes et al. [5] examined how varying levels of textual integration within charts, such as placement and semantic levels, affect interpretation. Their findings suggest that readers generally prefer some level of explanatory text alongside data visualizations, even if it increases visual complexity.

Importantly, Stokes et al. also compared graphical visualizations with text-only formats and concluded that text-only representations should not be dismissed. While they may lack the visual engagement of charts, text-based formats can offer clear and structured interpretations, especially for audiences less familiar with interpreting visual data. This supports the inclusion of a text-only condition in the present study to assess its comparative effectiveness in communicating key messages and data-driven insights.

3 Method

PSAs

This study utilized three real public service announcements (PSAs) from various campaigns as the foundation for the visual stimuli. One was selected from the U.S. Department of Agriculture [6] and the other two were selected from the Centers for Disease Control and Prevention (CDC) [2].

When selecting PSAs for inclusion in this study, several criteria were applied to ensure relevance, validity, and replicability. Each PSA was required to be an officially published work from a reputable government agency to ensure credibility and enhance the ecological validity of the research. Additionally, selected PSAs needed to be in infographic format and include some form of graphical data representation, aligning with the study's focus on data visualization. Finally, the underlying data had to be publicly available so that accurate, unembellished charts could be reproduced for the plain graph condition. These criteria ensured that the materials used were both authentic and reflected real-world communication practices.

Each PSA was adapted into three distinct conditions to systematically examine the impact of different visualization formats:

1. **Chart**: A traditional graph representing the PSA's data in a simplified, unembellished format.
2. **Infographic**: A reproduction of the original PSA used in the campaign, preserving its visual design and stylistic elements.
3. **Text**: A written explanation of the data, presented without any visual components.

Each condition was designed using Figma [3], a collaborative web-based interface design tool commonly used for prototyping and visual design. Because the original fonts used in the PSAs were not publicly available, each infographic was recreated using a visually similar font to maintain the overall aesthetic. These recreated versions constitute the infographic condition in the study.

To create the plain chart condition, the original datasets were first used to generate charts in Microsoft Excel. The chart type for each PSA was selected to closely resemble the visual structure of the corresponding infographic, maintaining consistency in how the data was presented. These charts were then added to the Figma prototype.

For the text-only condition, each data point was translated into a corresponding textual description. In the cases of the asthma and fire budget PSAs, the number of data points was sufficient to allow a full textual representation without overwhelming the layout or readability. However, the cigarette PSA presented a challenge due to the density of data in its line graph. To address this, the text condition for that PSA was adapted to summarize the overall trend rather than listing each individual point. This summary was drawn directly from the original CDC publication, ensuring it both encapsulated the underlying data and conveyed the original PSA's intended message with fidelity.

Throughout all three conditions, visual consistency was maintained by standardizing the background color, title, subtitle, and font style across designs based on the original PSA. This controlled presentation allowed the study to isolate the effects of visual format on comprehension, recall, and preference.

Measurements

Comprehension

To evaluate participants' comprehension of each presented PSA, they were asked to answer three free-response questions following each viewing. One question was to report the overall message of the PSA, and the other questions were to provide two additional facts about the given data presented. Each response was independently assessed and scored by the researcher. Participants received 1 point for each correct answer and 0 points for incorrect or incomplete responses, resulting in a possible score of 0 to 3 for each PSA.

Visualization Literacy

To measure participants' ability to interpret visualizations, they completed the Mini-VLAT (Visualization Literacy Assessment Test), a standardized instrument developed to assess an individual's proficiency in understanding and analyzing graphical information [4]. Including this measure is critical, as individuals with lower levels of visualization literacy may encounter greater difficulty interpreting visualizations in PSAs, potentially limiting the effectiveness of the message.

Memory

Memory retention was evaluated using a similar approach to the comprehension assessment. Participants' recall responses were graded by the researcher, yielding a memory score for each PSA recalled. If a participant failed to recall a specific PSA, a default score of 0 was assigned to that PSA topic. This approach allowed for consistent scoring while capturing both the accuracy and completeness of participants' memories.

Preference

Participants were asked to rank the five PSA variations according to five subjective dimensions of preference:

- **Overall Preference:** Which visualization do you like best?
- **Memorability:** Which is the easiest to remember?
- **Effectiveness:** Which would best raise awareness about the issue in your community?
- **Trustworthiness:** Which one do you trust the most?
- **Attention:** Which captures your attention the most?

Several of these dimensions, namely overall preference, memorability, and attention, were adapted from prior work by Bateman et al. [1]. Additional dimensions were included to reflect key attributes of effective PSAs better. The dimension of *effectiveness* was included to evaluate how well a visualization could fulfill one of the core objectives of a PSA: raising public awareness. Meanwhile, *trustworthiness* was added to assess participants' perceptions of the visualizations as credible and reliable sources of information.

Demographic

To account for potential differences in the interpretation of visualizations due to background factors, demographic information was collected. Participants reported their age, education level, frequency of reading or creating charts, and general preference for learning through pictures or text. These variables provided context for analyzing individual differences in comprehension, memory, and preference responses.

Participants

A total of 48 participants were recruited using convenience sampling. All participants completed an online survey administered through Qualtrics. All responses were anonymous. The survey took around 20-25 minutes to complete. Recruitment was conducted through informal channels, including direct messages to friends, Instagram stories, and Snapchat. Participants were asked to complete the survey in one sitting to control for the time interval between the presentation of PSAs and the recall task.

According to the demographic survey, the majority of participants were between 18 and 24 years old, indicating a predominantly college-aged sample. Most reported having some prior experience both reading and creating charts, suggesting a baseline familiarity with data visualizations.

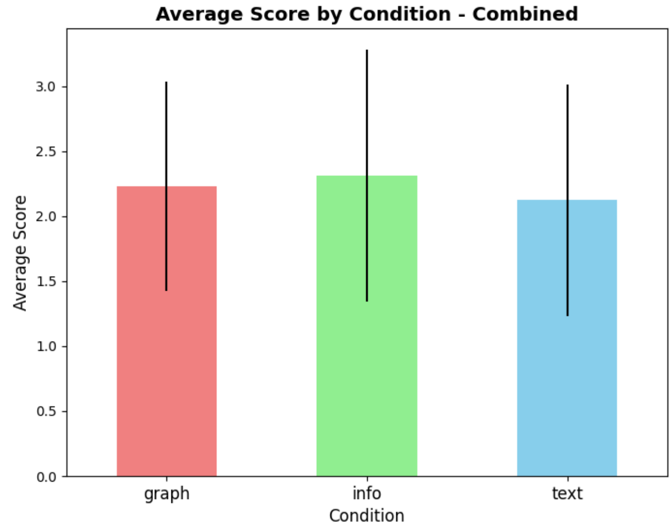


Fig. 6: Average accuracy for each condition

Procedure

Participants began the study by viewing one randomly assigned condition of each of the three PSA stimuli. After viewing each visualization, they were asked to complete a set of comprehension questions designed to assess their understanding of both the intended message and the underlying data:

1. What is the main message of this visualization?
2. List two other facts about it.

The goal of these questions was to evaluate participants' understanding of the key message conveyed by the PSA and their ability to extract supporting details from the data.

Following the comprehension task, participants completed the Mini-VLAT, a short, standardized assessment of visualization literacy designed to be completed in under five minutes [4]. This task also served as a temporal buffer before the memory assessment.

After completing the Mini-VLAT, participants were asked to recall as many of the previously viewed visualizations as they could. For each visualization they remembered, they responded to the following free-response questions:

1. What was the topic of the visualization?
2. Describe the visualization (e.g., type of visualization you saw, colors that were used)
3. What was the main message of the visualization?
4. Enter a fact you remember about the data in the visualization
5. Enter another fact you remember about the data in the visualization

Questions 3-5 mirrored those from the initial comprehension task, enabling comparison between immediate understanding and later recall. Question 1 was used to match each response with the corresponding PSA stimulus.

After the recall task, all three variations of the same PSA were presented simultaneously. Participants were asked to rank the visualizations according to the five definitions of preference mentioned earlier. This was done for each of the 3 PSA stimuli. Finally, participants completed a short demographic questionnaire, as mentioned earlier. This information was collected to contextualize participants' performance and preferences across conditions.

4 Results

Accuracy

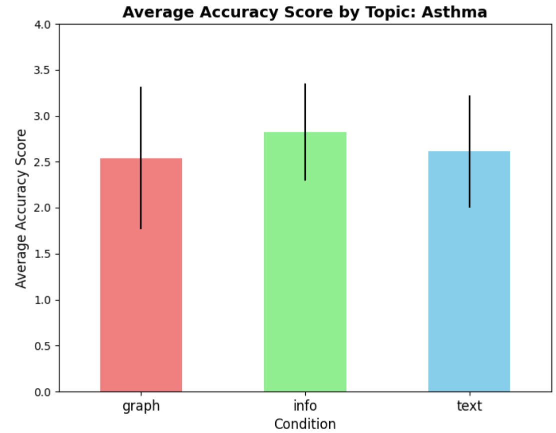
We start by analyzing how well participants understood each PSA based on their accuracy scores. As mentioned in section 3.2, participants were assigned an accuracy score of 0-3 points based on how many comprehension questions they answered correctly. Figure 6 presents the mean accuracy scores for each condition. Participants in the infographic condition achieved an average accuracy score of 2.31 (SE = 0.97, n = 48), followed by the graph condition with a mean of 2.23 (SE = 0.81, n = 48), and the text condition with a mean of 2.13 (SE = 0.98, n = 48). Since each participant was exposed to all three design types, a Friedman test was conducted to assess whether there were statistically significant differences in comprehension across conditions. The results ($X^2(2)$ = 1.79, p = .409) indicate no significant difference in accuracy across visualization formats. These findings suggest that participants performed similarly across all three conditions, with no single design yielding significantly higher comprehension scores.

When analyzing accuracy scores by PSA topic (Figure 7), a visual inspection of the results reveals consistent trends for the asthma (Figure 7a) and fire (Figure 7c) stimuli: participants performed best with the infographic condition, followed by the text, and then the graph. In contrast, the cigarette (Figure 7b) PSA displayed a reversed pattern, with the graph condition yielding the highest accuracy, followed by text, and then infographic.

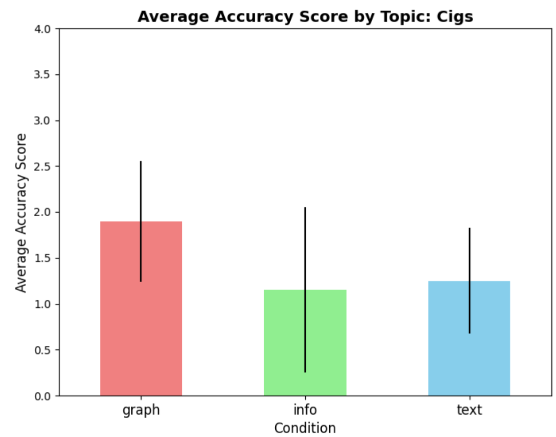
Because each participant was exposed to only one version of each topic, a Kruskal-Wallis test was conducted to compare the three independent groups per topic. For the asthma stimulus, the test revealed no significant difference in accuracy scores across conditions, $H(2) = 2.192$, $p = .334$. Similarly, the *fire* stimulus showed no significant effect of design type, $H(2) = 0.815$, $p = .665$. These findings suggest that, for these two topics, the PSA design format did not significantly affect participant comprehension.

However, a significant difference was observed for the cigarette PSA, $H(2) = 9.425$, $p = .009$, indicating that the design format did influence accuracy. Analysis using pairwise Mann-Whitney U tests revealed a significant difference between the graph and text conditions ($U = 77.5$, $p = .007$). This suggests that the graph format of the cigarette PSA led to higher comprehension than the text version, highlighting the influence of visual design on participants' performance on this topic.

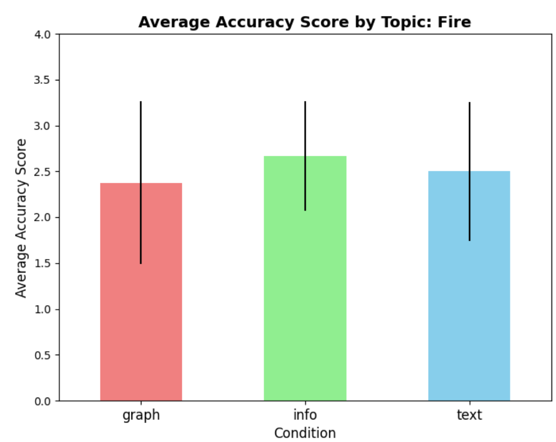
Interestingly, the cigarette PSA yielded lower overall accuracy across all conditions than the asthma and fire PSAs. One possible explanation for this pattern is that many participants included responses that, while relevant to the PSA's general theme, were not grounded in the data and were therefore scored as incorrect. For instance, several participants reported that "smoking is bad" and that "smoking causes heart disease", information that is present in the infographic's subtitle. This behavior suggests that participants may have struggled to extract meaning from the data, potentially due to a lack of interpretable visual features. As a result, they may have redirected their attention to more immediately accessible textual elements, such as headlines or subtitles. This pattern underscores the importance of clear data presentation and may reflect a breakdown in information hierarchy, where peripheral content distracts from the primary data message. It also highlights the importance of a good balance between the amount of text and the amount of data present.



(a) Average accuracy score for each condition given the asthma PSA



(b) Average accuracy score for each condition given the cigs PSA



(c) Average accuracy score for each condition given the fire PSA

Fig. 7: Average accuracy score broken down by PSA stimuli

Visualization literacy

To assess participants' visualization literacy, each participant completed the Mini-VLAT. The total number of correct responses served as each participant's visualization literacy score, with a maximum possible score of 12. The average Mini-VLAT score across all participants was 9.77.

To explore the relationship between visualization literacy and comprehension, a Spearman correlation was conducted between Mini-VLAT scores and accuracy scores for each response (Figure 8). The analysis revealed no significant association (Spearman's ρ = 0.05, p = .53), indicating that visualization literacy did not have a measurable effect on participants' ability to accurately interpret the PSAs across design conditions.

Memory

To evaluate how well participants retained information from the PSAs, accuracy scores from the initial comprehension task were compared to memory scores collected during the recall phase. The overall average drop from comprehension to recall was most significant in the graph condition (0.72), followed by the infographic (0.66), and lowest in the text condition (0.53). When broken down by topic (Figure 9),

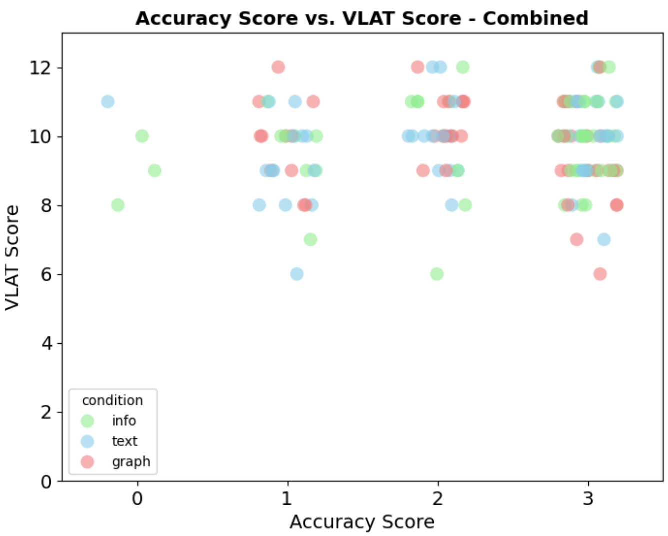


Fig. 8: Accuracy score vs. VLAT score

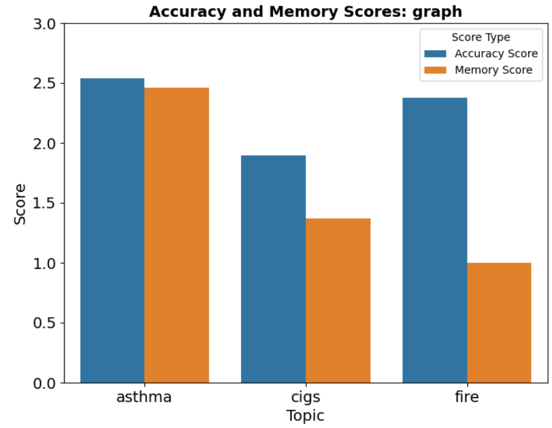
memory scores were consistently lower than initial accuracy scores across all conditions and PSA topics. The most significant reduction was observed in the graph condition of the fire PSA (-1.375), while the smallest reduction occurred in the infographic condition of the asthma PSA (-0.08). Overall, the average decline in score across all conditions was -0.606, indicating a measurable loss in information retention.

An interesting trend emerges when comparing comprehension and memory performance by topic. In terms of accuracy, participants consistently performed best on the asthma PSA, followed by fire, and worst on cigarettes. One might expect this pattern to persist in the memory scores; however, it was observed only in the infographic condition (Figure 9b). In the text condition (Figure 9c), participants performed slightly better on the fire PSA than on asthma. In the graph condition (Figure 9a, the asthma PSA still resulted in the highest memory scores, followed by cigarettes, with fire performing the worst.

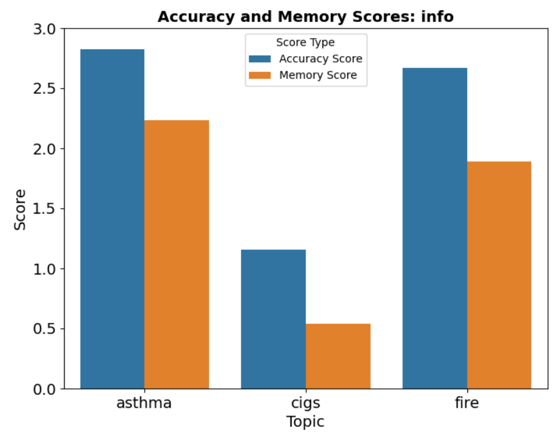
Upon examining open-ended responses from the comprehension task, several participants exposed to the graph condition of the fire PSA indicated uncertainty about what the visualizations conveyed, even when they provided correct answers. This lack of confidence may have negatively impacted their ability to recall the content later. In contrast, participants who viewed the infographic condition may have benefited from visual cues. For example, the "burning trees" illustration in the fire PSA offered semantic context and may have aided memory retention despite any initial confusion. This interpretation aligns with findings by Bateman et al. [1], who argued that visual embellishments can enhance memorability by reinforcing associations between imagery and content. Such embellishments may have mitigated uncertainty and contributed to more resilient recall, potentially explaining why the graph condition for the fire PSA showed the most significant decline in memory score.

Preference

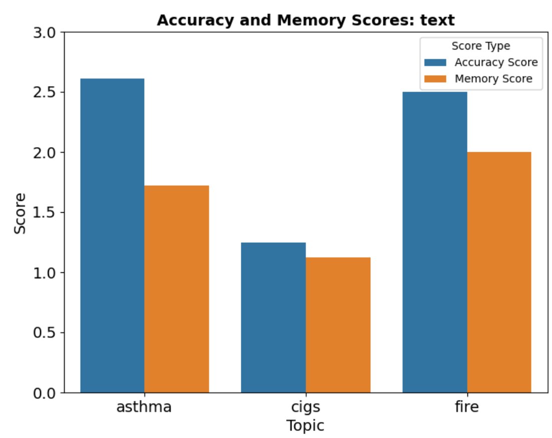
To evaluate participants' subjective preferences for different visualization formats, we analyzed their rankings across five definitions: overall preference, attention-capturing ability, awareness-raising potential, memorability, and perceived trustworthiness. Each participant ranked the three formats for each definition. Lower ranks indicate greater preference. Table 1 presents the mean rankings, the results from Friedman tests, and the results of a pairwise Wilcoxon signed-rank test assessing differences across conditions and PSA topics. In all five definitions, the Friedman tests revealed statistically significant differences among the visualization formats (all $p < .01$). This implies that the rankings for each condition were not random and that there is a statistically significant difference in how participants ranked the different visualizations for each definition. Participants did not rate all



(a) Average accuracy and memory score for the graph condition



(b) Average accuracy and memory score for the infographic condition



(c) Average accuracy and memory score for the text condition

Fig. 9: Average accuracy and memory scores by condition

visualization types equally and had clear preferences.

Infographics emerged as the most preferred format in four of the five definitions (Overall preference, attention, awareness, and memory). For these definitions, text was consistently ranked lowest, while graphs generally ranked in the middle. A pairwise comparison using Wilcoxon signed-rank tests (Table 1) confirmed significant differences between all format pairs across these four definitions (all $p < .01$).

However, a contrasting pattern emerged for the definition of trust. Graphs were rated as the most trustworthy, with an average ranking of 1.58, whereas both text and infographics were rated equally and

Treatment	Graph Ranking	Info Ranking	Text Ranking	Friedman Stat	p-value	Text/Info	Graph/Info	Graph/Text
Overall	1.92	1.55	2.53	71.51	< 0.01	< 0.01	< 0.01	< 0.01
Attention	2.10	1.10	2.81	212.17	< 0.01	< 0.01	< 0.01	< 0.01
Awareness	2.02	1.42	2.56	92.35	< 0.01	< 0.01	< 0.01	< 0.01
Memory	1.91	1.49	2.60	89.54	< 0.01	< 0.01	< 0.01	< 0.01
Trust	1.58	2.21	2.21	37.50	< 0.01	< **0.85**	< 0.01	< 0.01

Table 1: Average rankings for each design based on the definition of preference. A Friedman test was conducted to determine whether participants ranked conditions differently. A pairwise comparison using Wilcoxon signed-rank tests confirmed significant differences between all format pairs across all definitions except for the Text/Info pair for trust.

significantly less trustworthy, with average rankings of 2.21. The Friedman test again indicated a significant effect of visualization type ($p < .01$), and Wilcoxon comparisons revealed significant differences between graphs and both other formats ($p < .01$). Notably, Figure 1 shows that no significant difference was found between text and infographic in terms of trust ($p = .85$). This implies that graphs stood out in terms of trustworthiness, while participants didn't meaningfully differentiate between text and infographic formats.

5 Discussion

This study examined how three data visualization formats (infographic, text, and plain graph) affect comprehension, memory, and user preference in the context of infographic PSAs. Consistent with prior work by Bateman et al. [1], no statistically significant difference in comprehension was observed across conditions at the aggregate level. However, a closer examination of the cigarette PSA revealed that participants performed significantly better when viewing the graph condition compared to text and infographic. One explanation may be that the graph format offered clearer access to the underlying data and provided more details that were not present in the infographic and text variants. This suggests that in the absence of a strong data presence, participants may divert their attention to peripheral cues that do not directly support comprehension.

Furthermore, visualization literacy did not affect comprehension. This may be due to a ceiling effect, as the majority of participants demonstrated relatively high visualization literacy scores, potentially masking any moderating influence of this variable.

In terms of memory, we observed a decline in information retention across all three conditions, with the largest drop occurring in the graph condition, followed by the infographic, and the smallest drop in the text condition. This trend is somewhat counterintuitive, as one might expect infographics to yield superior memory performance given their ability to provide additional context about the data. A possible explanation is that, as seen with the fire PSA, the design of the infographic was confusing to many participants. This may have reduced their confidence in their response in the comprehension task and impeded encoding and recall. In contrast, the text condition, while less visually engaging, may have supported more straightforward interpretation and semantic encoding. Participants' confidence during comprehension may have played an important role in their ability to recall information during the memory task.

Preference data showed strong, consistent favorability toward infographics. Participants ranked them highest in overall preference, attention-grabbing ability, awareness-raising, and memorability. However, a clear divergence emerged regarding perceived trust, with participants rating traditional graphs as the most trustworthy format.

Since this was a pilot study, the sample size was relatively modest ($n = 48$) and consisted primarily of younger, educated individuals recruited through convenience sampling. As a result, the findings may not generalize to broader, more diverse populations typically targeted by PSAs, including individuals with lower levels of education or limited experience interpreting data visualizations. Moreover, the visual stimuli were not fully standardized across conditions. Differences in layout dimensions, font styles, and the amount of subtitle text may have introduced unintended variability, potentially influencing participants' attention and interpretation.

Future research should address these limitations by recruiting a more diverse participant pool and standardizing key design elements across PSAs, such as font style, dimensions, and the amount of supplementary text. Additionally, the present study was limited in its use of visualization types, focusing primarily on bar charts, stacked bar charts, and a line graph. Subsequent studies should incorporate a wider variety of chart forms to investigate whether specific graph types are more or less effective when embedded as embellishments within infographic designs. Building on the work of Stokes et al. [5], it would also be valuable to systematically vary the quantity and placement of textual annotations within infographics to assess their impact on comprehension, recall, and subjective preference. Finally, future work should refine and expand the definition of "preference" to better understand how audiences interpret and engage with PSAs.

6 Conclusion

This study examined how infographic, graph, and text formats influence comprehension, memory, and preference in public service announcements. While comprehension scores did not differ significantly overall, a topic-level investigation suggests that clarity and data density in visual design can meaningfully impact understanding. Interestingly, memory scores declined most for graph designs, potentially due to a lack of confidence in participants' responses during the comprehension task. Infographics were preferred across most dimensions, including attention, awareness, and memorability, but graphs were viewed as the most trustworthy.

Taken together, these findings highlight the importance of aligning visual design choices with a PSA's specific communication goals. Designers must balance clarity, engagement, and credibility, particularly when working with diverse audiences. While infographics may be good in attracting attention and aiding recall, graphs remain a critical tool for conveying trust. Textual elements, though less favored overall, may offer valuable support for comprehension and should not be dismissed.

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